

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Storypoints)

Date	1 June 2026
Team ID	SWTID-2026-8236
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Member
Sprint-1	Project Planning	USN-1	Create ER Diagram and Project Workflow	2	High	Srikanth
Sprint-1	Project Planning	USN-2	Collect GitHub Repository and Demo Links	1	High	Srikanth
Sprint-1	Data Collection	USN-3	Download Credit Card Approval Dataset	2	High	Srikanth
Sprint-1	Data Analysis	USN-4	Import Libraries and Read Dataset	2	High	Lakshmi
Sprint-2	Data Analysis	USN-5	Perform Univariate Analysis	2	Medium	Lakshmi
Sprint-2	Data Analysis	USN-6	Perform Multivariate Analysis	2	Medium	Lakshmi
Sprint-2	Data Analysis	USN-7	Generate Descriptive Statistics	1	Medium	Lakshmi
Sprint-2	Data Preprocessing	USN-8	Remove Duplicate Records	2	High	Naga Jyothi
Sprint-3	Data Preprocessing	USN-9	Handle Missing Values	2	High	Naga Jyothi
Sprint-3	Data Preprocessing	USN-10	Data Cleaning and Feature Engineering	3	High	Naga Jyothi
Sprint-3	Data Preprocessing	USN-11	Encode Categorical Variables	2	High	Mandakini

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Member
Sprint-3	Model Building	USN-12	Train Logistic Regression Model	2	High	Mandakini
Sprint-4	Model Building	USN-13	Train Decision Tree Model	2	Medium	Mandakini
Sprint-4	Model Building	USN-14	Train Random Forest Model	3	High	Mandakini
Sprint-4	Model Building	USN-15	Compare Models and Save Best Model	2	High	Mandakini
Sprint-5	Application Development	USN-16	Develop HTML User Interface	2	High	Srikanth
Sprint-5	Application Development	USN-17	Develop Flask Backend	3	High	Naga Jyothi
Sprint-5	Application Development	USN-18	Integrate Trained Model with Flask	2	High	Mandakini
Sprint-5	Testing & Deployment	USN-19	Test Application and Fix Bugs	2	High	Entire Team
Sprint-5	Documentation	USN-20	Prepare README, Report, Presentation and Conclusion	2	High	Entire Team

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	18	6 Days	07 Jul 2026	12 Jul 2026	18	12 Jul 2026
Sprint-2	20	6 Days	13 Jul 2026	18 Jul 2026	20	18 Jul 2026
Sprint-3	22	6 Days	20 Jul 2026	25 Jul 2026	22	25 Jul 2026
Sprint-4	20	6 Days	27 Jul 2026	01 Aug 2026	20	01 Aug 2026
Sprint-5	20	6 Days	03 Aug 2026	08 Aug 2026	20	08 Aug 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

$$AV=20/6=3.33 \text{ Story Points/Day}$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

Burndown Chart Data

Sprint	Planned Story Points Remaining	Actual Story Points Remaining
Start	100	100
Sprint 1	80	82
Sprint 2	60	62
Sprint 3	40	38
Sprint 4	20	18
Sprint 5	0	0